Math 150 – Exam 1 – Chapters 1, 2 – Spring 2015

I pledge to demonstrate personal and academic integrity in all matters with my fellow students, the faculty, and the administration here at USD. I promise to be honest and accountable for my actions; and to uphold the stature of scholastic honesty to better myself and those around me. Initials: __________

Name: Solutions ______________________ Section: 1 2

Instructions: **Read all directions carefully. Ask for clarification if you do not understand the directions. Show your work when necessary. Complete sentences and justifications are only necessary when indicated.**

There are 100 points on this test and 4 Extra Credit points.

Num.	Score	Out of
1		5
2		20
3		8
4		10
5		10
6		15
7		10
8		10
9		12
B		*5*
Total		100

Please notify me if you find errors in the solutions.

1. Let's get off to a good start. There are no wrong answers here. Write a complete sentence describing your your greatest mathematical strength.

2. Evaluate each limit.

(a) $\lim_{x \to 2^-} \frac{1}{x-2} =$ $-\infty =$

(b) $\lim_{x \to 2^+} \frac{1}{x-2} =$ ∞

(c) $\lim_{x \to 2} \frac{1}{x-2} =$ DNE

(d) $\lim_{x \to 1} \frac{3x^4 - 2x^3 + 6}{x^2 + 3} =$ $\frac{3-2+6}{1+3} = \frac{7}{4}$

(e) $\lim_{x \to 1} \frac{x^2 - 3x + 2}{x - 1} =$ $\lim_{x \to 1} \frac{(x-2)\cancel{(x-1)}}{\cancel{(x-1)}} = \lim_{x \to 1} x-2$

(f) $\lim_{x \to \infty} \sqrt{x^2-1} - x =$ $\lim_{x \to \infty} \left(\sqrt{x^2-1} - x\right)\frac{\left(\sqrt{x^2-1} + x\right)}{\left(\sqrt{x^2-1} + x\right)} = \lim_{x \to \infty} \frac{x^2-1-x^2}{\sqrt{x^2-1}+x}$

$\lim_{x \to \infty} \frac{-1}{\sqrt{x^2-1}+x} = 0$

3. Let f be a function. Suppose $f'(a)$ is positive. Circle all the true statements.

   (a) The slope of f is increasing at a.

   (b) The function f is increasing at a.

   (c) The function f is concave up at a.

   (d) The function is continuous at a.

4. Suppose $f(x)$ is continuous at $x = 2$, $f(2) = 9$, and $\lim_{x\to 4} g(x) = 2$.
   Compute $\lim_{x\to 4}[3f(g(x)) - x^2] =$

$= 3\lim_{x\to 4} f(g(x)) - \lim_{x\to 4} x^2$

$= 3f(\lim_{x\to 4} g(x)) - \lim_{x\to 4} x^2$

$= 3f(2) - 16$

$= 3(9) - 16$

$= 27 - 16$

$\boxed{= 11}$

5. Use the Intermediate Value Theorem to determine if there is a root of $2x^3 + x^2 = -2$ in the interval $(-2, -1)$.

$f(x) = 2x^3 + x^2$

$f(-2) = 2(-8) + 4 = -12 < -2$

$f(-1) = 2(-1) + 1 = -1 > -2$

Therefor by IMV theorem
there is a c in $(-2, -1)$
Such that
$2c^3 + c^2 = -2$

6. Recall the following two (equivalent) definitions of the derivative of a function f at a number $x = a$, denoted by $f'(a)$, are

Definition 1

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Definition 2

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

if this limit exists.

(a) Given $f(x) = -2x^2 + 10$ and $x = 1$. Use either definition 1 or 2 to find $f'(x)$. You must show all your steps, but no explanation of steps are necessary.

$$f'(1) = \lim_{x \to 1} \frac{-2x^2 + 10 - (\overset{8}{-2+10})}{x-1}$$

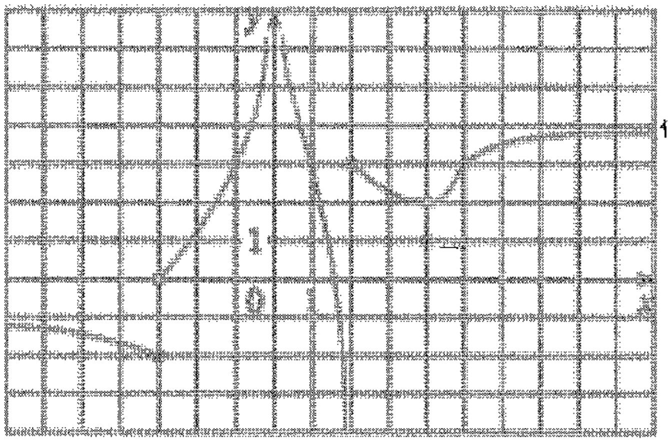
$$= \lim_{x \to 1} \frac{-2x^2 + 2}{x-1} = \lim_{x \to 1} \frac{-2(x^2-1)}{x-1}$$

$$= \lim_{x \to 1} -2(x+1) = -2(2) = \boxed{-4}$$

(b) Write a complete sentence that describes the geometric meaning of $f'(1)$.

The rate of change of the function f at $x = 1$ is -4.

7. List at least one vertical asymptote and at least one horizontal asymptotes (there are two of each type) of the following function $f(x)$ and use the definition of the type of asymptote to state why there is an asymptote:



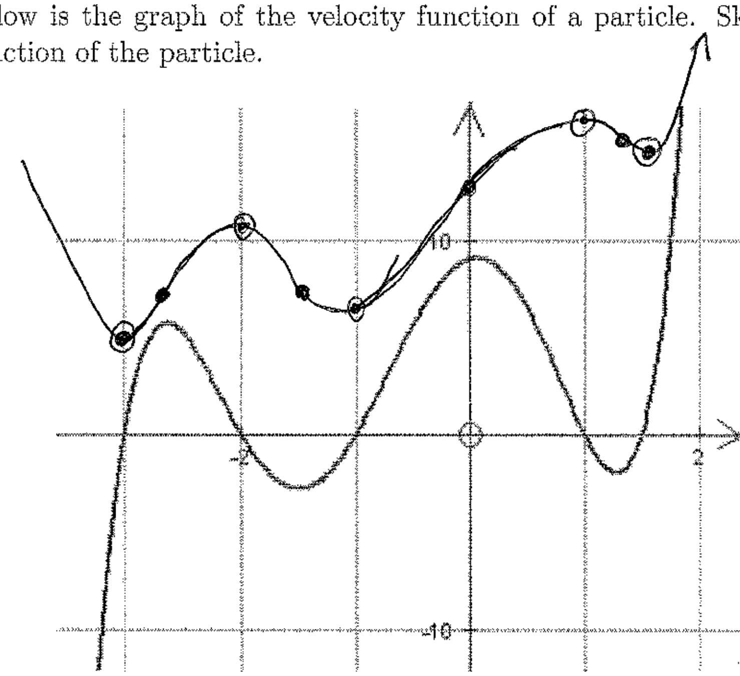
VA: $x=0$ and $x=2$

b/c $\lim_{x\to 0} f(x)=\infty$ b/c $\lim_{x\to 2^-} f(x)=-\infty$

HA: $\lim_{x\to\infty} f(x)=4$ so $y=4$ is a HA

$\lim_{x\to-\infty} f(x)=-1$ so $y=-1$ is a HA.

8. Below is the graph of the velocity function of a particle. Sketch the graph of a position function of the particle.



Answers will vary but abs max, min ⊙ & inflection pts • are ~~[illegible]~~ at given x values

9. Let

$$f(x) = \begin{cases} x^2\cos(\frac{4}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Is $f(x)$ continuous at $x = 0$?

$f(0)=0$, $\lim_{x\to 0} f(x) = \lim_{x\to 0} x^2\cos(4/x)$.

$-x^2 \leq x^2\cos 4/x \leq x^2$

$\lim_{x\to 0} -x^2 = \lim_{x\to 0} x^2 = 0$

so $\lim_{x\to 0} x^2\cos 4/x = 0$

so $f(0) = \lim_{x\to 0} f(x)$ so $f(x)$ is continuous at $x=0$.

10 **Bonus Question: This is for extra credit**
Let if $f(x) = \sqrt{x} - x^2$. Apply Definition 1 from Question 6 to find the derivative $f'(x)$.

$$\lim_{h\to 0} \frac{\sqrt{x+h} - (x+h)^2 - (\sqrt{x} - x^2)}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} - \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}\left(\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}\right) - \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h\to 0} \frac{x+h-x}{h(\sqrt{x+h}+\sqrt{x})} - \lim_{h\to 0} \frac{2xh+h^2}{h}$$

$$= \lim_{h\to 0} \frac{1}{\sqrt{x+h}+\sqrt{x}} - \lim_{h\to 0} 2x+h$$

$$= \frac{1}{\sqrt{x+0}+\sqrt{x}} - 2x + 0 = \frac{1}{2\sqrt{x}} - 2x.$$